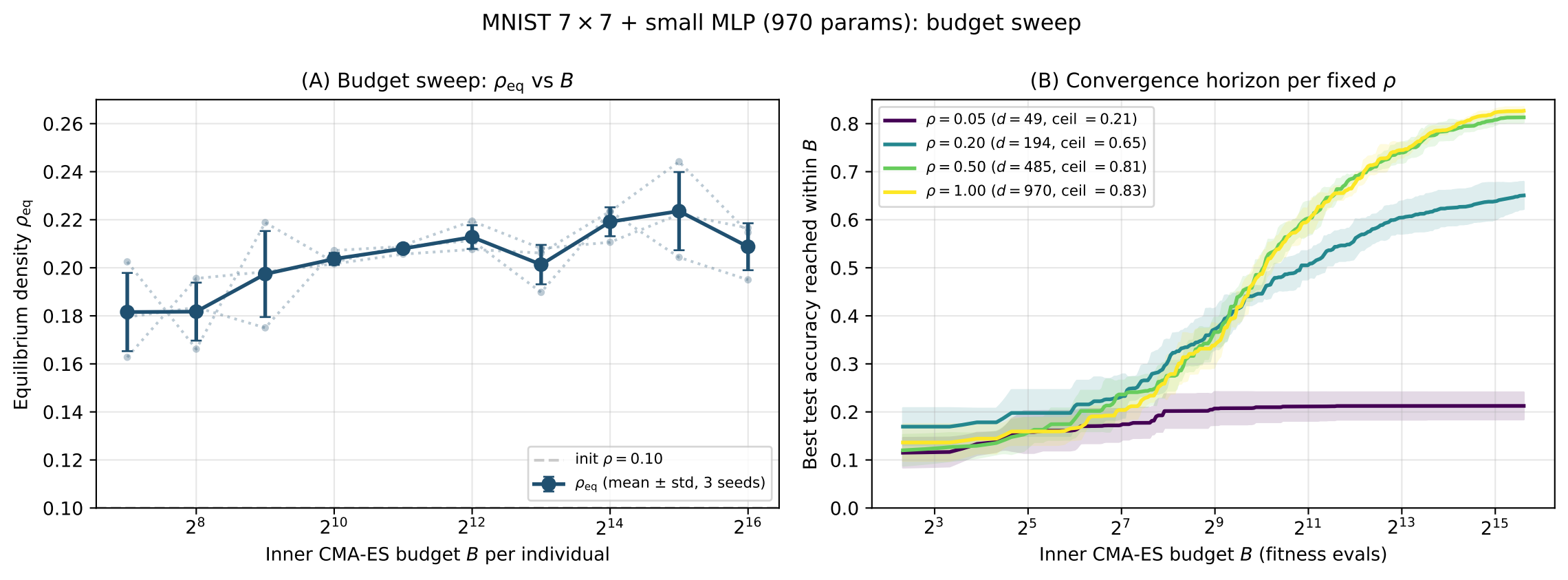
MNIST 7 × 7 + small MLP (970 params): budget sweep
(A) Budget sweep: ρ_{eq} vs B
Equilibrium density ρ_{eq}
0.26
0.24
0.22
0.20
0.18
0.16
0.14
0.12
0.10
2^{8}
2^{10}
2^{12}
2^{14}
2^{16}
Inner CMA-ES budget B per individual
init $\rho = 0.10$
ρ_{eq} (mean ± std, 3 seeds)
(B) Convergence horizon per fixed ρ
Best test accuracy reached within B
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
2^{3}
2^{5}
2^{7}
2^{9}
2^{11}
2^{13}
2^{15}
Inner CMA-ES budget B (fitness evals)
$\rho = 0.05$ ($d = 49$, ceil $= 0.21$)
$\rho = 0.20$ ($d = 194$, ceil $= 0.65$)
$\rho = 0.50$ ($d = 485$, ceil $= 0.81$)
$\rho = 1.00$ ($d = 970$, ceil $= 0.83$)